

Literature Benchmark Brief: Sootweg_2003_TPWRS

Title: General model for variable speed wind turbines including a pitch control model

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Venue: IEEE Transactions on Power Systems | **DOI:** 10.1109/TPWRS.2002.805018

Cluster: Cluster C: Aerodynamics & Control Cliff

Core Theoretical Mechanism & Model Formulation

Full-order aeroelastic $C_p(\lambda, \beta)$ formulation and first-order pitch actuator dynamics

Empirical Scale & Experimental Dataset

Wind turbine grid disturbance and fault-ride-through field tests

Relevance to IEEE TSTE & Sustainable Energy Systems

Foundational mathematical model for variable-speed turbine aerodynamics and pitch control

Identified Gaps & Operational Vulnerabilities

Static polynomial approximation exhibits numerical singularities at extreme pitch angles

Role in Manuscript Defense & Reviewer #2 Proof

Mathematical source for Eq. (2)-(4) and physical basis of rated wind speed derivative cliff